What's new

The Bourgain Project

Terence Tao · 19 October 2026

https://terrytao.wordpress.com/2026/10/19/the-bourgain-project/

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[This is a guest post by Lukas Liehr, João Pedro Ramos and Mitchell Taylor – T.]

We are launching the Bourgain Project, a community-driven initiative to revisit, expand upon, and explore the body of work of Jean Bourgain, one of the most prolific mathematicians of his time. The goal of this initiative is to develop a deep, comprehensive understanding of each one of Bourgain's over 500 individual papers, while discussing and connecting the common threads and unifying ideas that run throughout his research.

Bourgain made numerous contributions across many fields, such as harmonic analysis, number theory, partial differential equations, and functional analysis. However, many of his papers are notoriously difficult to grasp. Our initiative aims to make his ideas more accessible to the broader mathematical community.

In addition to digesting Bourgain's contributions, another main goal of the project is to explore new formats for presenting mathematics beyond the traditional journal article. We encourage contributors to share digestions and expositions of his papers in whatever form, to collaborate in groups, and to engage in discussions on our community Zulip channel.

For more information, we invite anyone interested to visit our website at https://bourgain-pedia.com/, and check out our latest article digestions.

Comments (7)



igorrivin · 19 October 2026

By an interesting coincidence, I have been doing something along these lines, and the results are not quite what I would have hoped for. Check out https://mathreferee.dimensionreducers.ai



· 19 October 2026

Igor Rivin, always trying to attach himself to the next big thing. I guess this is epsilon less degrading than managing a Russian oligarch's money…



igorrivin · 20 October 2026

Thank you, Anonymous Coward, for your slanderous statements. I wish I did manage a Russian oligarch's money, though.

 **Sam Hopkins** · 20 October 2026

There have been many questions on MathOverflow over the years of the form “I am trying to understand a technical step in a paper of Bourgain…”: see the comments to https://mathoverflow.net/questions/121751

 **boldlya2e7c26b8c** · 20 October 2026

https://www.ams.org/journals/notices/202011/rnoti-p1716.pdf

 · 20 October 2026

Can I just suggest to find another name instead of “digestion”? Because we all know what is the end product of a digestion, maybe we could find some better naming. Besides jokes, this seems a great project honestly.

 **el houcein** · 29 October 2026

I’m posting here my email on Zulip, hopefully, it will reach more colleagues :

“Dear All
I would be happy to start a collaboration on digesting the following paper of Bourgain: J. Bourgain, *Double recurrence and almost sure convergence*, J. Reine Angew. Math. 404 (1990), pp. 140-161.

This is one of Bourgain’s more difficult papers. A useful companion paper is available here: https://arxiv.org/pdf/2012.06323. THis later paper was revised after un e-exchange of email with Prof. Terrence Tao!

I would like to specify that this digestion should be carried out in the spirit of the corollary of my recent paper: https://arxiv.org/abs/2610.30797. In particular, we may need to formalize Bourgain’s paper in Lean.

I hope to hear from any colleagues who are interested in the project.

With best regards,
el-Houcein”